Bishnu Prasad Kar

Bhubaneswar, India | +91 7750023564 | karbishnu2430@gmail.com
linkedin.com/in/bishnu-prasad-kar | github.com/Bishnu2430 | bishnuprasadkar.vercel.app

ABOUT

AI/ML engineer and Computer Science undergraduate who builds complete systems and measures whether they work: OCR and retrieval pipelines with reported accuracy, LLM output gated by validators and red-team tests, and simulations checked against real observations. Works in Python, Rust and TypeScript, from model training to APIs and GPU code.

EDUCATION

B.Tech in Computer Science and Engineering Expected Aug 2027
Gandhi Institute of Excellent Technocrats (GIET), Bhubaneswar *CGPA 8.8/10 (through semester 6)*
Coursework: Machine Learning, Artificial Intelligence, Data Structures and Algorithms, Operating Systems, DBMS

EXPERIENCE

Data Science Intern Dec 2025 – Feb 2026
Jyesta Corporate Entity × E-Cell, IIT Roorkee
- Built loan-prediction and recommendation systems: feature engineering, several classifiers, model explainability.

Generative AI and Data Analytics Trainee 2025 – 2026
Central Tool Room & Training Centre (CTTC), Ministry of MSME *Bhubaneswar*
- Generative AI (2026: LLMs, LangChain, RAG, end-to-end capstone); Data Analytics (2025: Python, SQL, Tableau).

PROJECTS

Nabz – Lab Reports Explained | *Python, FastAPI, PostgreSQL, pgvector, React, TypeScript, Three.js, Docker* 2026
- Full-stack app that reads a photo or PDF of a blood-test report and explains it in three languages on a 3D body map.
- Extraction pipeline (text layer, RapidOCR with deskew, layout-agnostic parser, LOINC mapping over 108 tests): 100% field accuracy on digital PDFs, 95.5%+ on simulated phone photos; 99.7% of rows mapped to the right test.
- Grounded LLM explanations (RAG over NLM MedlinePlus) pass a deterministic validator and an LLM judge, else fall back to a template; all 139 red-team cases pass in CI.
- Trend engine (Theil–Sen, Mann–Kendall, RCV) confirms 88% of planted drifts at five visits, at 1.0% false alarms.

Worldline – Relativistic Universe Sandbox | *Rust, wgpu, egui* Oct 2026 – present
- Physics engine that simulates the real solar system from NASA JPL data (459 moons, 28,331 belt objects) with relativistic gravity and an adaptive IAS15 integrator; Earth stays within 0.41 km of JPL after a year.
- Every model ships with a validation test against theory or observation, 65+ passing: Mercury's precession 42.9807″ vs. 42.9805″ per century; a GW150914 replay leaves 63.0 $M_\odot$ (LIGO: 62 ± 4).
- GPU ray-traced black-hole shadows; SPH fluid planets torn apart by tides at 8,144 particles and 30+ FPS.

Q-CONSENSUS – Multi-Agent LLM Reasoning | *Python, FastAPI, llama.cpp, Qiskit Aer, Solidity, Docker* 2026
- Five agents debate on a local LLM; a QAOA solver over quantum-kernel similarity picks the consensus answer, benchmarked against a classical baseline.
- SHA-256 hash-chained events, anchored on a private Ethereum chain, make every run replayable and verifiable.

Misinformation Vaccine – Fake-News Detection | *PyTorch, Transformers, scikit-learn, FastAPI, PostgreSQL* 2025
- Fine-tuned DistilBERT on 43,944 articles to 0.9995 F1, against a 98.84%-accuracy TF-IDF baseline; documented the dataset bias behind the near-ceiling score.
- Served it as a URL-to-verdict FastAPI service with scraping and stored history.

TECHNICAL SKILLS

Languages: Python, Rust, TypeScript, SQL, Bash
Machine Learning: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, SHAP, NLP, computer vision, OCR
Generative AI: LLMs, RAG, agentic systems, LangChain, FAISS, pgvector, llama.cpp, Ollama, LLM guardrails
Systems: FastAPI, PostgreSQL, Docker, Git, GitHub Actions, Linux, React, Spark, Hadoop, Qiskit

PUBLICATIONS

- *Entropy-Optimized Adaptive Hybrid Thresholding for Retinal Blood Vessel Segmentation*. SPARK-2K26, 2026.
- *CityFlow: A Scalable Big Data Framework for Real-Time Analytics in Computing 4.0*. NCCENGT-2025.
- *A Modular and Scalable Arithmetic and Logical Unit Design through Multiplexer-Based Controller*. ICSTRE-2024.